Quang Hoang

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U.S. Citizen | Active Public Trust | Eagle Scout | Community Service Awards

Summary

Software Engineer specializing in **Python**, **AWS infrastructure**, **RAG pipelines**, and **agentic workflows**

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL, C, Java, HTML/CSS, Power Fx
Frameworks: LangChain/LangGraph, DeepEval, LiveKit, Django, Flask, PyTorch, Hugging Face Transformers, pytest, unittest
Developer Tools: Git/GitHub/GitLab, Amazon Bedrock, Terraform, FAISS, Docker, Kubernetes, LiveKit, Jira, uv
Libraries: Deepgram, Boto3, Azure OpenAI

Experience

Custom Software Engineering Senior Analyst — Oct 2026 – Present
Accenture Federal Services — *Washington, DC*

- Work with the *Department of Education* on building an Agentic IVR for *Federal Student Aid*

AI Engineer — Apr 2026 – Oct 2026
Lumiture, LLC — *Washington, DC*

- Fixed 7 QA-reported LLM voice agent defects in production by rewriting **LangGraph** system prompts
- Scaled voice regression coverage to 408 cases via a **LiveKit** + **DeepEval** LLM-judge pipeline on **Amazon Bedrock**
- Drove team decision to migrate to **Deepgram** Flux TTS from Aura-2 via a 30-category **A/B** evaluation
- Found a chunking strategy raising **RAG** Recall@5 from 0.867 to 0.923 by benchmarking 39 **FAISS** configs with **RAGAS**

AI Trainer — Dec 2025 – Apr 2026
Handshake AI Fellowship — *Remote*

- Rated ~1,000 multimodal LLM responses on 1-5 rubrics, writing comparative justifications for training
- Supported prompt development and evaluation through independent research

Agile Software Engineer (Academic Capstone) — Aug 2025 – Dec 2025
William & Mary — *Williamsburg, VA*

- Resolved 9 sprint issues on a 6-person Agile team, including 6 error-handling and crash fixes in **Jira**
- Shipped 11 reviewed PRs to a team GitHub repo using feature-branch workflows
- Extended a **Dockerized Django** app that replaced manual Qualtrics CSV exports with charted survey analysis
- Protected data-parsing views from regressions by writing 23 Django unit tests

Software Developer Intern — May 2025 – Aug 2025
Soft Tech Consulting — *Fairfax, VA*

- Automated internal knowledge retrieval by deploying a Copilot Studio LLM chatbot with OneDrive RAG
- Streamlined contract management by modeling CLIN-task order dependencies in a **Power Fx** dataverse schema

Projects

Static Site Generator | *Python, HTML/CSS, CI/CD* — May 2025 – Jun 2025

- Automated web deployment into 1-step Markdown-to-HTML build pipeline by building static site generator in **Python**
- Ensured HTML generator stability and parser reliability by writing 100 unit tests via **unittest**

Generative AI & NLP Engineering Portfolio | *PyTorch, Hugging Face, Pandas, OpenAI* — Jan 2025 – May 2025

- Reached 0.88 **CodeBLEU** predicting masked Python if-conditions by fine-tuning CodeT5-Small on 50K functions
- Benchmarked 5 prompting strategies on 4 LLMs, including GPT-4.1 and Codestral, across 22 code tasks
- Built a Java code completion 3-gram model in Python with ~177 validation perplexity, best of n = 3, 5, 9

Education

William & Mary — Williamsburg, VA
Bachelor of Science in Computer Science | *Major GPA: 3.49* — *Aug 2022 – Jan 2026*